Sheaf-cohomology QC for volume-EM registration: pre-registered negative results and an exact view-gauge invariance

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Abstract

We operationalized cohomological inconsistency (failure of adjacent-frame displacement measurements to form a 1-coboundary on the coverage nerve) as a quality-control detector for volume-electron-microscopy registration, and evaluated it, pre-registered, on three real channels: two distant ROIs of an EMPIAR-10478 HeLa SBF-SEM block and one vendor-registered EMPIAR-10070 FIB-SEM muscle stack. Three findings survive. (1) An **exact view-gauge invariance** (an elementary cancellation identity, stated as a remark below): whole-view displacement is an exact gauge invariance of relative-content measurement; the contribution is empirical — a shift-channel anomaly scores AUROC exactly 0.500 at every amplitude on real phase-correlation measurements. (2) **Detectable bias is empirically detector-saturated within the tested closure-residual family**: an injected edge defect crosses AUROC 0.9 only at ∼4–5× the triangle-closure noise scale on raw SBF (GLR-optimal and max-abs detectors differ by <0.03; a re-implemented pairwise photometric QA baseline scores *lower* than the sheaf channel at matched relative scale), while pre-registered FIB-SEM residuals sit ∼15× lower in absolute pixels yet still admit no $\leq 3\times$ detection. closure-σ scales span two orders of magnitude across techniques, a threshold sweep finds the photogrammetry prune rule budget-driven rather than rule-driven (workable τ: 2.5× SBF vs 3× FIB-SEM at 1% false flags; scoped to single-edge defects at K=6, global bundle adjustment untested), and a stitcher-free control with pixel-exact known truth reproduces the same saturation (AUROC ≤ 0.9 through 5×; view-desync exactly gauge) with detectability invariant to graph density $K \in \{5, 7\}$ — so QC thresholds must be absolute per-task tolerances, not scale-free noise multiples, and the saturation is measurement-physics, not stitching artifact. (3) An **instrument-disclosure protocol**: a self-referential-anchor defect that flat-lined the first detector at chance was caught by an algebraic gauge proof before any gate result was seen. All results, including nulls, are frozen in git before each run; code and artifacts are public at https://github.com/ScivizLabVienna/e3-sheaf-data.

1 Introduction

Large volume-EM reconstructions are assembled by registering thousands of adjacent section images; registration errors propagate silently into segmentation and connectomics. Existing automated quality assessment for this step is learned and pairwise — optical-flow SIQA metrics [4, 3], learned features [5], vision-language judges [6] — and carries no structural statement about *which* defects a given statistic can or cannot see.

Cellular sheaves on coverage nerves offer exactly such a statement: displacement measurements on overlaps are a 1-cochain, a consistent registration is a coboundary, and the closure residual of every triangle is a per-edge-usable inconsistency signal with no learned parameters. Prior registration work uses the same formalism to *eliminate* loop error globally (Poisson/Hodge solving) [2], never to *detect* per-edge defects against a noise-calibrated threshold.

We adopted the minimal instance of this detector, pre-registered acceptance gates, and ran it on real archives. The headline outcome is negative and useful: within the tested statistics, what the detector can see is set by the channel, not by detector cleverness (we do not rule out statistics outside the closure-residual family), and the two structural claims below are exact.

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2 Formal setting

Frames $0, \ldots, N-1$ form a path nerve; each adjacent pair $(j, j+1)$ carries a measured translation $m_{jj+1} \in \mathbb{R}^2$ (phase correlation on overlapped ROIs). A *closure window* of K consecutive edges forms triangles whose residual is $\rho = m_{12} + m_{23} - m_{13}$ (global-reference legs measured independently at wide baseline). Two defect classes: **M_edge** — a wrong offset on one edge (e.g. a bad correspondence patch); **N_shift** — a whole-view translation desync.

> **Remark (view-gauge invariance).** *Adding a per-view translation t_j to all measurements touching view j leaves every closure residual invariant; hence no coboundary-channel statistic can discriminate an N_shift at any amplitude.* This is an elementary cancellation identity, not a new structural result; its value here is that it predicts and explains an exactly-0.500 empirical null, which no learned metric can.

Proof: residuals are alternating sums in which each t_j appears with coefficients summing to zero. The empirical instance is exact, not approximate (§4).

> **Proposition (redundancy dilution).** *In a window of K edges solved by weighted least squares, one corrupted edge's offset contributes to per-row residuals at $O(1/K)$; detection difficulty grows with the very redundancy that improves fit quality.*

3 Methods (frozen before each run)

Instrument (E6b): independent global-reference legs by wide-baseline (+30-frame) phase correlation plus adjacent lag-10 rows; triangle-closure residuals define σ_c per channel, emitted *before* any injection trial. Trials: $K = 6$, 40 trials per class, defect levels $\{0.5, 1, 2, 3, 5\} \times \sigma_c$; detectors max-abs residual and GLR-optimal whitened χ^2 energy [1]. Gate (identical for E6/E7/E8/E9): AUROC ≥ 0.9 at level $\leq 3 \times \sigma_c$, else detector-saturation verdict for that channel. Channels: EMPIAR-10478 SBF ROI1 (tile on disk 604,039,195 B, 151×2000×2000), ROI2 (740,048,035 B), EMPIAR-10070 FIB-SEM MITOMAP stack [7] (777,443,899 B, 291×1595×1675, vendor TTrAC-registered). A synthetic campaign (P1–E5, oracle conditions) closed the space before real data.

4 Results

Table 1: Closure σ_c and GLR AUROC at $\{0.5, 1, 2, 3, 5\} \times \sigma_c$ (bootstrap CIs in artifacts). N_shift: 0.500 exactly at all levels (ROI1).

Channel	σ_c (px)	0.5×	1×	2×	3×	5×
SBF ROI1	1.78 / 1.66	0.600	0.720	0.844	0.890	0.943
SBF ROI2	1.46 / 1.47	0.593	0.696	0.823	0.881	0.927
FIB-SEM	0.11 / 0.14	0.571	0.586	0.628	0.666	0.737

Table 4 and Figure 4 report the level curves. No channel meets the gate at $\leq 3\times$; both SBF ROIs cross 0.9 only between 3× and 5× (E8 replicates E6/E7 on a distant ROI). GLR energy beats max-abs by $\leq +0.025$ AUROC everywhere; a re-implemented pairwise photometric QA baseline (EMSIQA/OFSIQA recipe [4, 3]; 360 clean ROI1 pairs) scores *worse*, AUROC 0.53 at 3×. Detector design adds little *within the tested family*; statistics outside it remain open. FIB-SEM inverts the mechanism — vendor pre-registration drives σ_c two orders of magnitude down, and redundancy dilution (Proposition) plus absolute-scale mismatch leave AUROC <0.9 even at 5×. *Closure-σ self-calibration does not normalize detection across channels.*

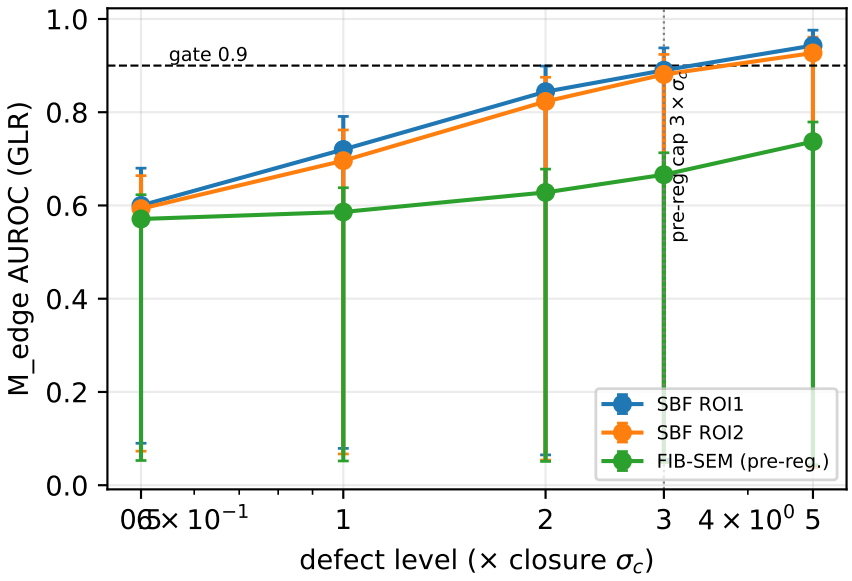


Figure 1: MDB level curves for the injected M_edge defect on all three channels (GLR detector; markers from the committed result artifacts, error bars are the pre-registration-frozen bootstrap CIs). The frozen gate (0.9 at $\leq 3\times$) is missed everywhere; SBF crosses between $3\times$ and $5\times$, FIB-SEM has not reached 0.9 by $5\times$.

5 Discussion

Operational rule. The field practice of pruning registration residuals above 2–3×RMS [10] implicitly assumes noise-dominated channels. Our threshold sweep (GLR energy, hard flag at $\tau\times$null-RMS, $\tau \in \{1.5, \ldots, 4\}$, 86/120/226 calibration windows per channel) shows the practice is *budget-driven, not rule-driven*: a 2–3× prune catches 53–74% of injected $\leq 3\times$ SBF defects but only at a 0.8–4.7% false-flag rate (SBF defect residuals sit in the far chi^2 tail; null RMS badly underestimates it), while the same rule flags only 19–30% of FIB-SEM defects at a comparable budget and needs $\tau\approx 1.5\times$ just to reach 13% false flags versus $\tau\geq 2.5\times$ on SBF: no universal τ transfers, and the operative dial is the tolerated false-flag rate, not a noise multiple. Thresholds must be absolute per-task tolerances. The same caveat applies to calibration of learned QA metrics [4, 6].

Disclosure protocol. The first E6 instrument contained a self-referential-anchor defect that made the tested statistic an exact gauge by construction (AUROC flat 0.500). It was diagnosed algebraically and disclosed *before* any gate verdict; the rebuilt instrument (independent global legs) reproduced theory. We report this as a methodological contribution for agent-run science.

Limitations. One FIB-SEM stack (no ROI variety); no independent ground truth on EMPIAR-10070 (shifts baked by NEWSTACK; extended header byte-verified history-only), so the oracle-ceiling comparison was run only on SBF legs; serial-section TEM and array tomography untested. σ_c on the FIB-SEM channel may partly reflect TTrAC smoothing (single stack, n=1): the stitcher-free known-truth control (E12) removes stitching as a confounder for saturation, but acquisition-side smoothing per se remains unresolved — that channel's $\sigma_c\approx 0.1$ px sits below the ~ 0.9 px phase-correlation floor measured on SBF texture under known truth, which we cannot fully account for.

6 Related work

Hodge/Poisson elimination of loop error in global registration [2]; EM stitching QA: EMSIQA [4], OF-SIQA/DEMIS [3], learned features [5], VLM judges [6]; sheaf-inconsistency mathematics: consistency ra-

dius [8]; streaming H^1 [9]. None evaluates coboundary residuals as calibrated per-edge QC, and none reports pre-registered negative results for volume EM.

7 Reproducibility

Git-campaign record (every gate dated before its run), frozen evaluators, result artifacts and bootstrap CIs: `https://github.com/ScivizLabVienna/e3-sheaf-data`; source data by byte-size-verified EMPIAR accessions. Rebuild: `e6b_measure.py` → `e7_glr.py` (numpy/scipy/tifffile).

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